Foundations of Operations Research

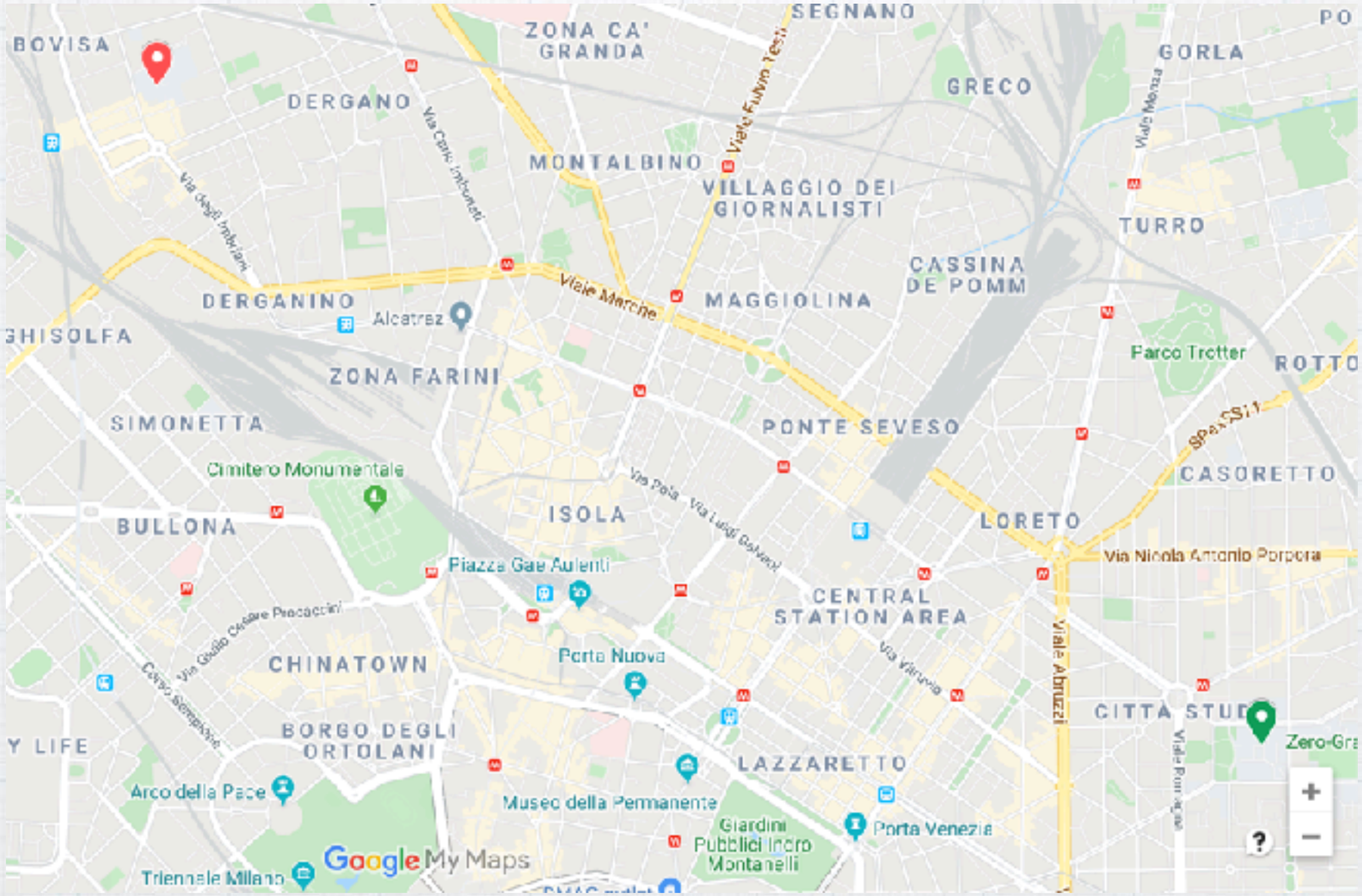
Federico Malucelli (lectures + ex)
Nicolò Gusmeroli (ex + lab)

Today's challenges

* Figure out what "operations research" is about
* Our teaching approach
* Organization of the course
* Instructions:
  * material
  * exam rules



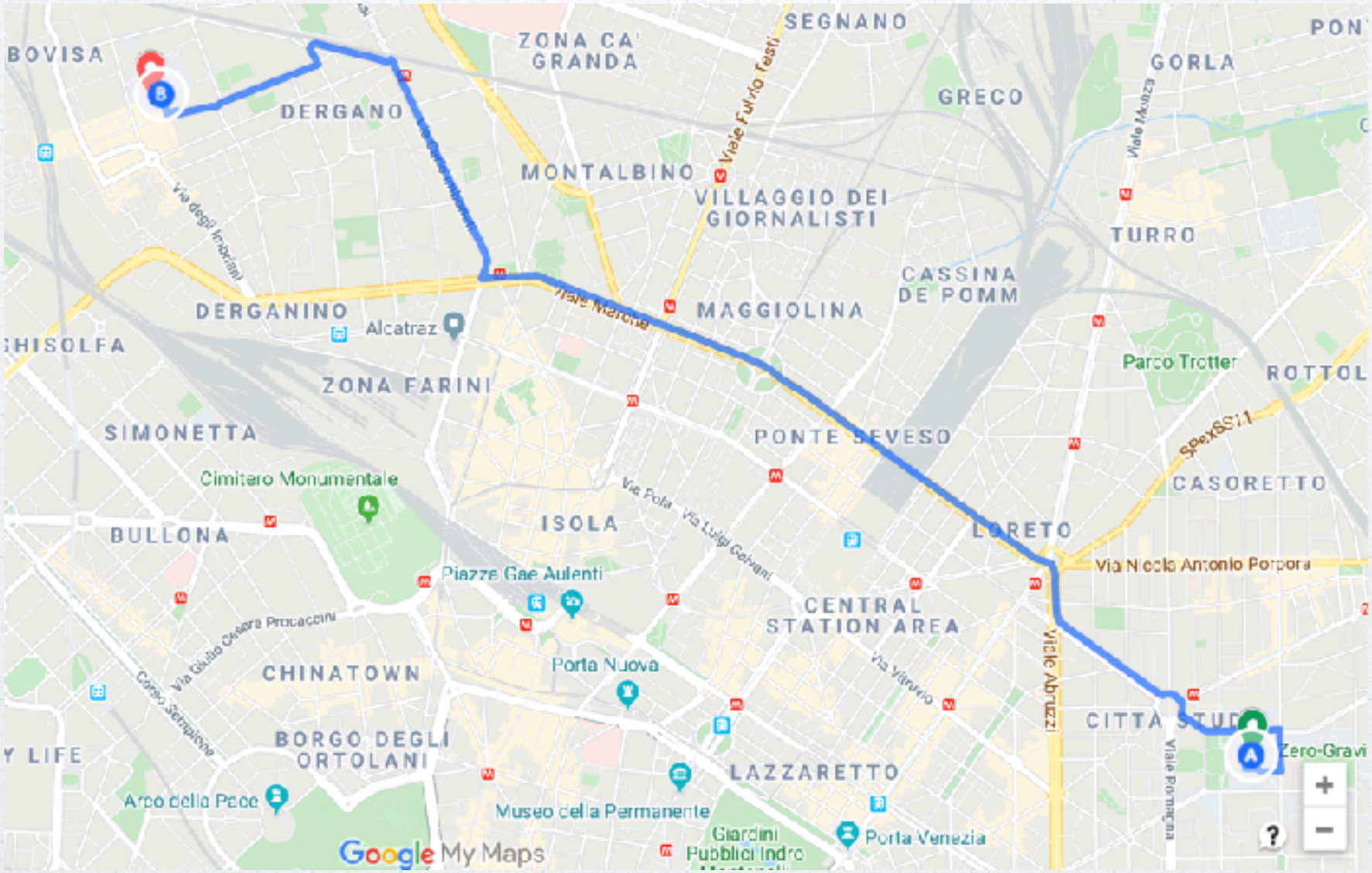
What is Operations Research (OR)?



Find the "best" itinerary from an origin to a destination

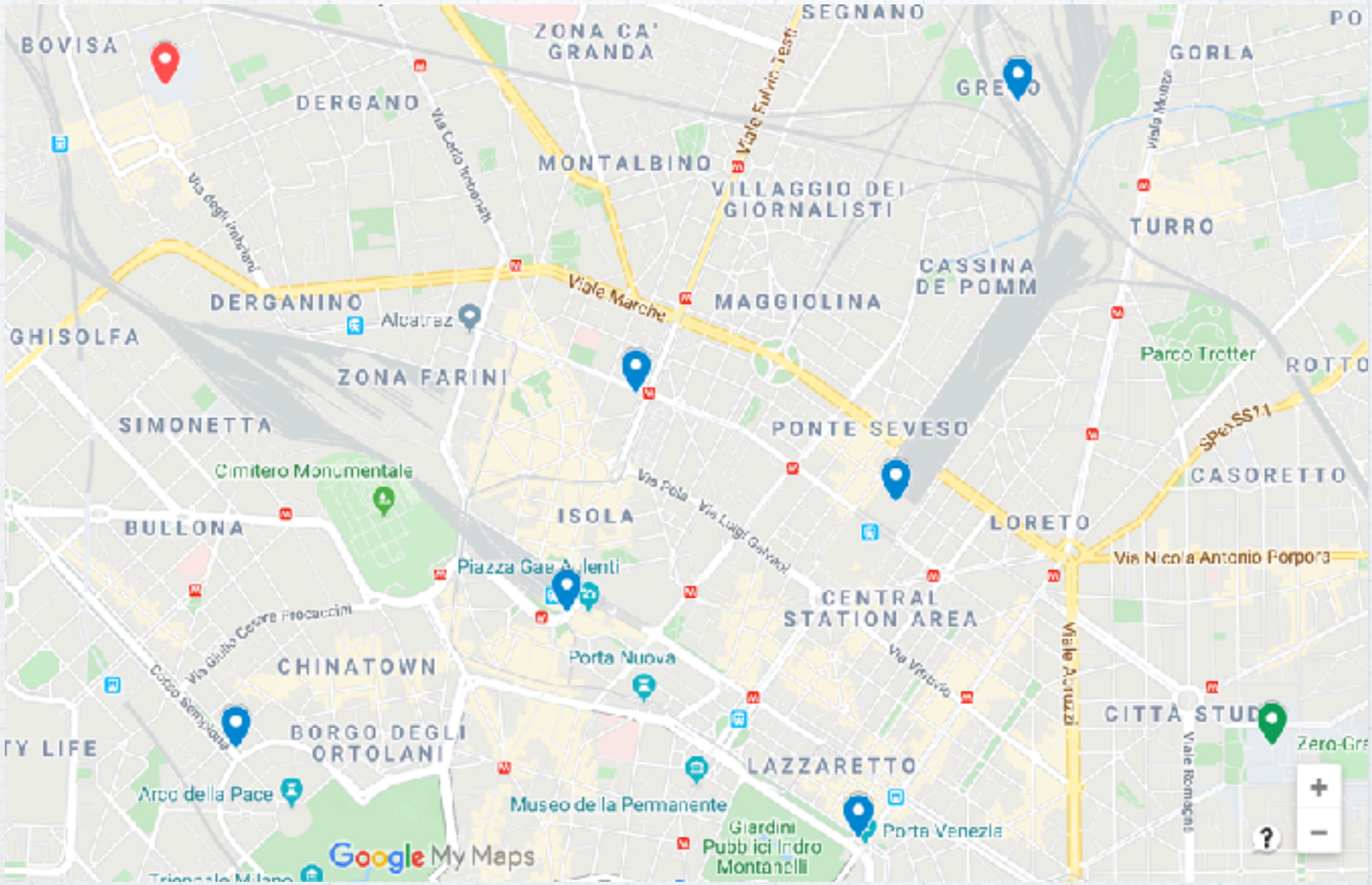
Criteria: shortest distance, minimum time, ...

What is Operations Research (OR)?



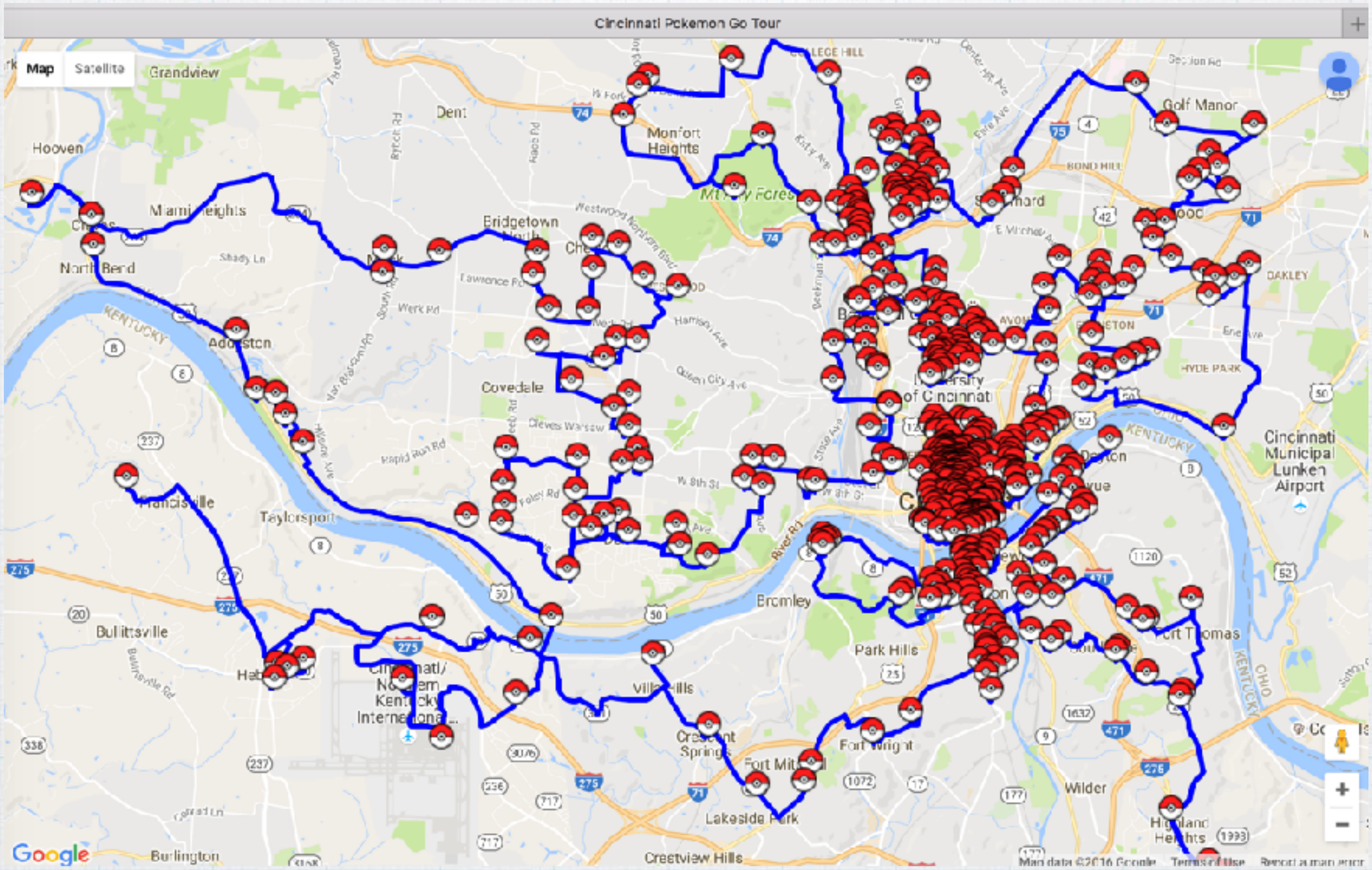
Travel time dependent on time

What is Operations Research (OR)?



Additional set of intermediate stops

Pokemon go



I want to catch them all in the minimum time

What is Operations Research (OR)?





* set up a team buying real players following some rules (number of backwards, forwards, etc.)
* limited budget of fantasy money
* each week depending on the players stats of the last game played a score is attributed to each team
* maximize the score of the team

Can OR help in dealing with global warming?

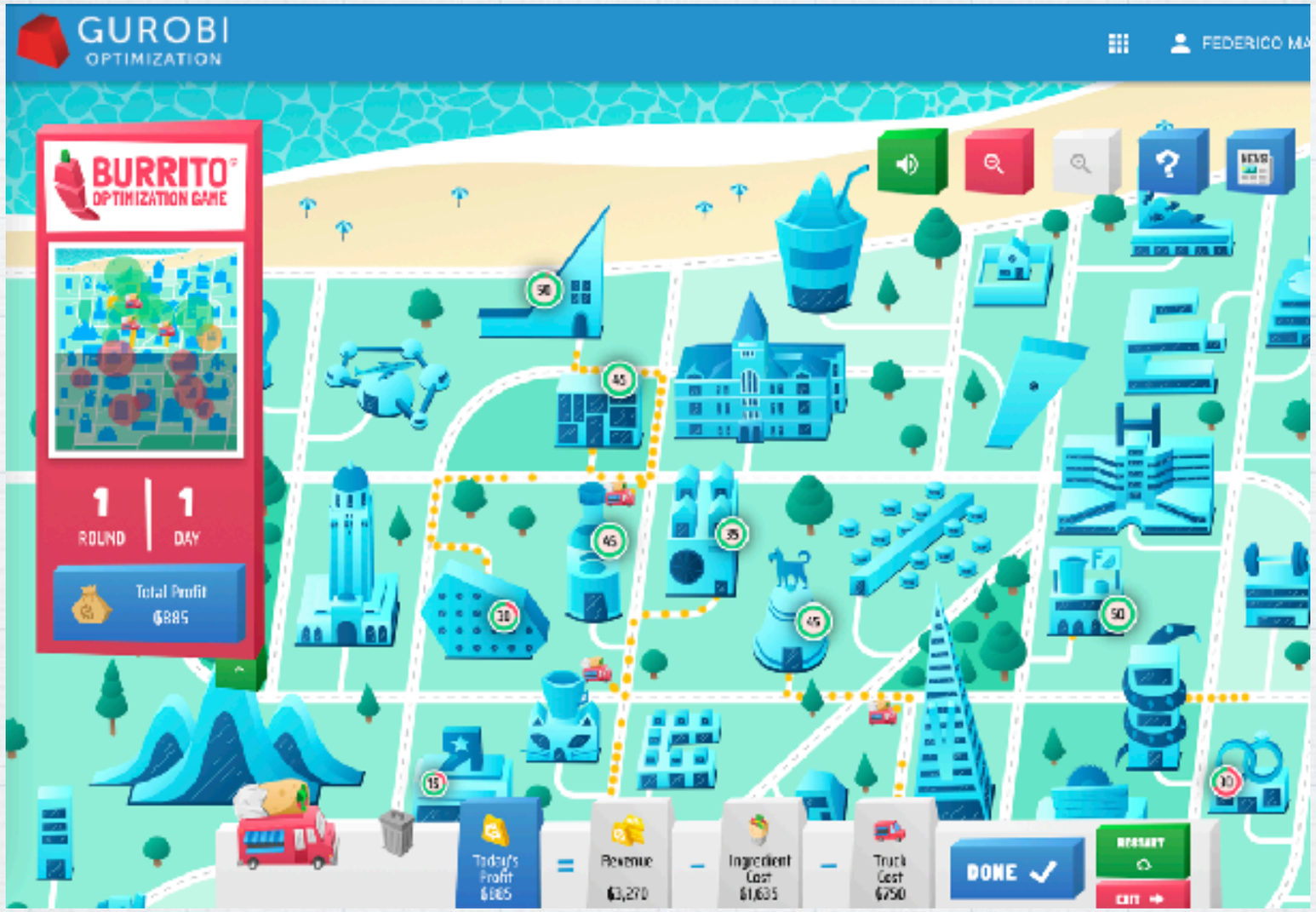


video

https://www.youtube.com/watch?v=dxaHMUqdPuA

article

An optimization game



game

https://burrito.gurobi.com/burrito-game/map?scenario=7PrneevPXP1qd6F9NK10

What is Operations Research (OR)?

Support in Decision Making in any complex system

HOW?

Mathematical models
Quantitative methods
Efficient algorithms

Maths

Computer Science

Analytics

Use the science to help in taking decisions

What kind of decisions?

* **Strategic level:**
  e.g. design a railroad
* **Tactical level:**
  e.g. how many trains to use
* **Operational level:**
  e.g. schedule the trains
* **Real time level:**
  e.g. disruption management

At any level from top management to everyday problems

OR was born during II World War

Optimization is as old as humanity





Queen Dido and the foundation of Carthage (814 BC)

Isoperimetric problem

Fields medal 2018
A. Figalli



Practical cases of OR application

* Why did you enroll in this course?
* What route did you follow to reach Politecnico?
* Where do you eat and what do you eat?

Probably you do not apply consciously OR methods
but you solve OR problems

If you want to discover more about OR

* http://www.scienceofbetter.org/
  classification of success applications
* http://www.informs.org/
  American OR society
  Franz Edelman Award record
* http://www.airo.org/
  Italian OR association

How to learn OR

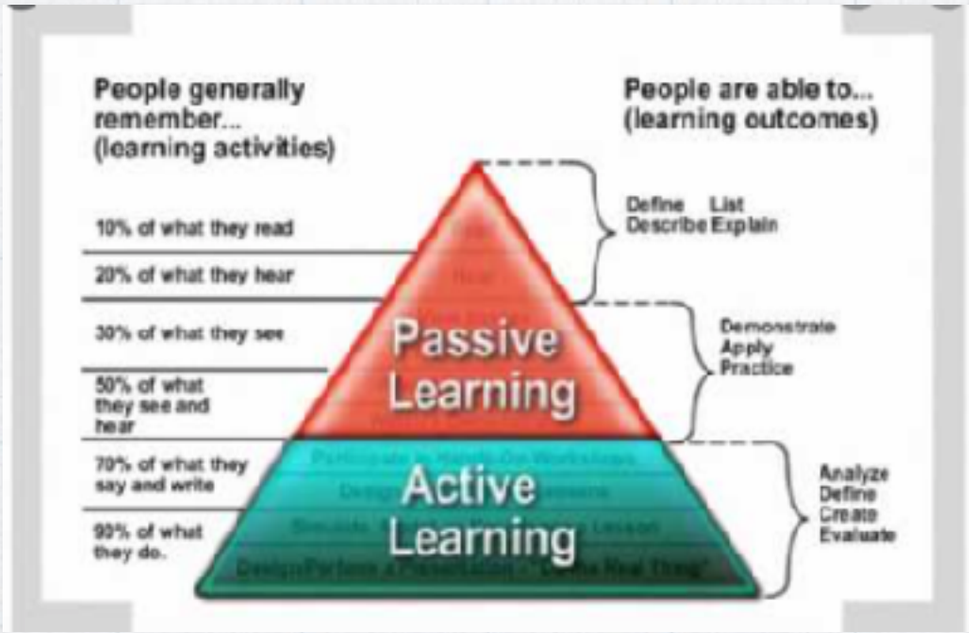
Confucius (551 bC - 471 bC)

"I hear and I forget

I see and I remember

I do and I understand"

"active learning"



https://www.tes.com/lessons/PdLpPcrW1lzRPA/active-learning

One important advice

OR is easier to learn if you put your hands on the problems



Scope of the activity:

- get acquainted with the "active learning mood"
- have a better understanding of optimization

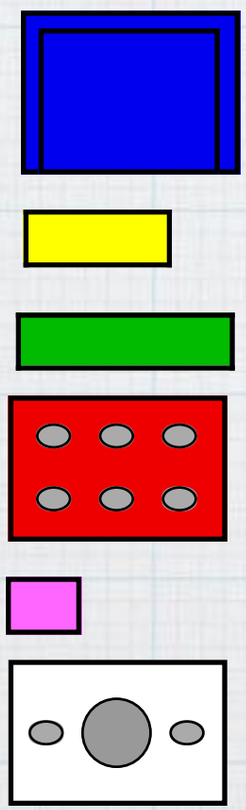
Webeep: Materials/Material for the activities in class

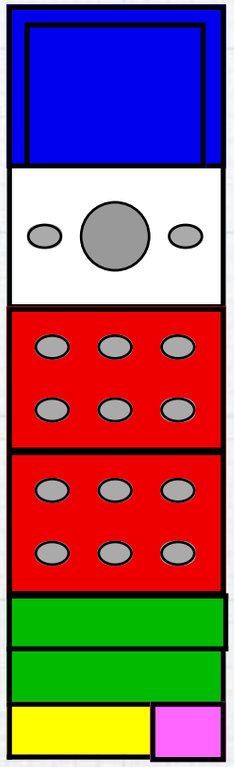
phones.pdf

A simple assembly problem



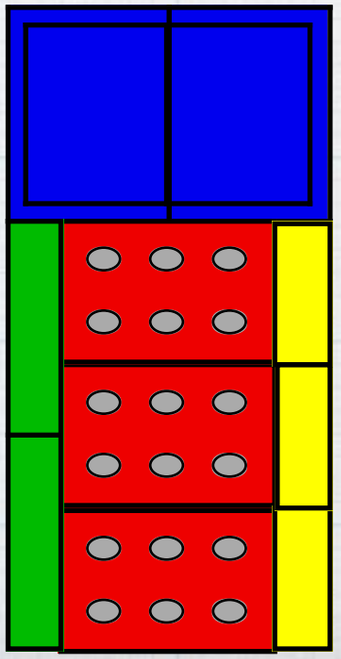
components





Phone A

Score A



Phone B

Score B

Objective:
Maximize the overall score

Solution

* Are you sure that the solution that you found is the **best** one?
* How can you **certify** it?
* How much are you willing to spend to have one more display, or keyboard, or memory board?
* Is it worth introducing a completely new product?

How OR approaches the problem

* **Decisions**
  What decisions did you take?
  How do you characterize your solution?
* **Rules**
  How do you combine components to obtain a product?
* **Objective**
  What do you have to optimize?

Decisions = variables

x_A = how many phones of type A

x_B = how many phones of type B

Rules = Constraints

The number of used components cannot exceed the available ones

$1x_A + 2x_B \leq 10$	display
$2x_A + 2x_B \leq 18$	memory
$1x_A + 3x_B \leq 12$	transmission
$2x_A + 3x_B \leq 21$	keyboard
$1x_A \leq 9$	navigation
$1x_A \leq 10$	camera
$x_A, x_B \geq 0$	non negativity

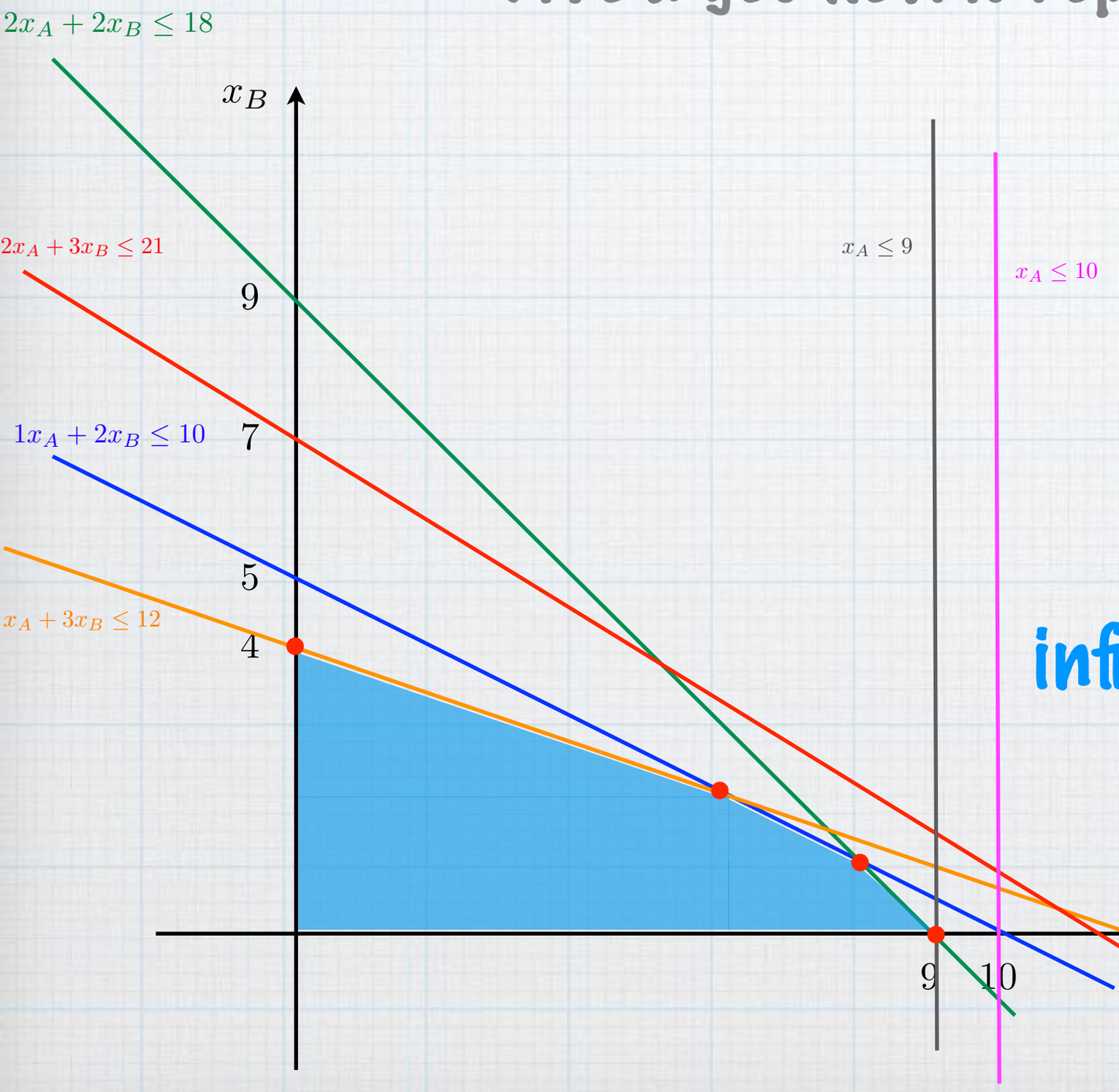
Objective function

Maximize the overall score

$$\max 3x_A + 8x_B$$

One simple solution method

Exploit the fact that we have only 2 variables
Give a geometric representation



"Interesting" solutions

$x_A = 0, x_B = 4 \quad v = 32$

$x_A = 6, x_B = 2 \quad v = 34$ ←

$x_A = 8, x_B = 1 \quad v = 32$

$x_A = 9, x_B = 0 \quad v = 27$

Feasible region
infinite number of solutions

Skills of an OR expert

* **Analyze the problem**
  Knowledge of the application field
  Communication skills to understand and explain
* **Knowledge of the OR methods**
* **Ability in building mathematical models**
* **Interpret the results**



Search on the internet information about the job of operations research

Program of the course

* **Introduction to OR**
  Optimization problems and their formulations; decision variables, objective function, constraints. Modeling techniques

* **Graphs and network flows problems**
  Trees and graphs. Shortest paths, maximum flow, minimum cost flow, assignments. Some data structures and solution algorithms. Complexity analysis

* **Linear Programming**
  Geometry of Linear Programming, vertices and basic solutions. Duality theory, complementary slackness, the simplex method; geometrical and economical interpretation. Basic solutions and optimality conditions.

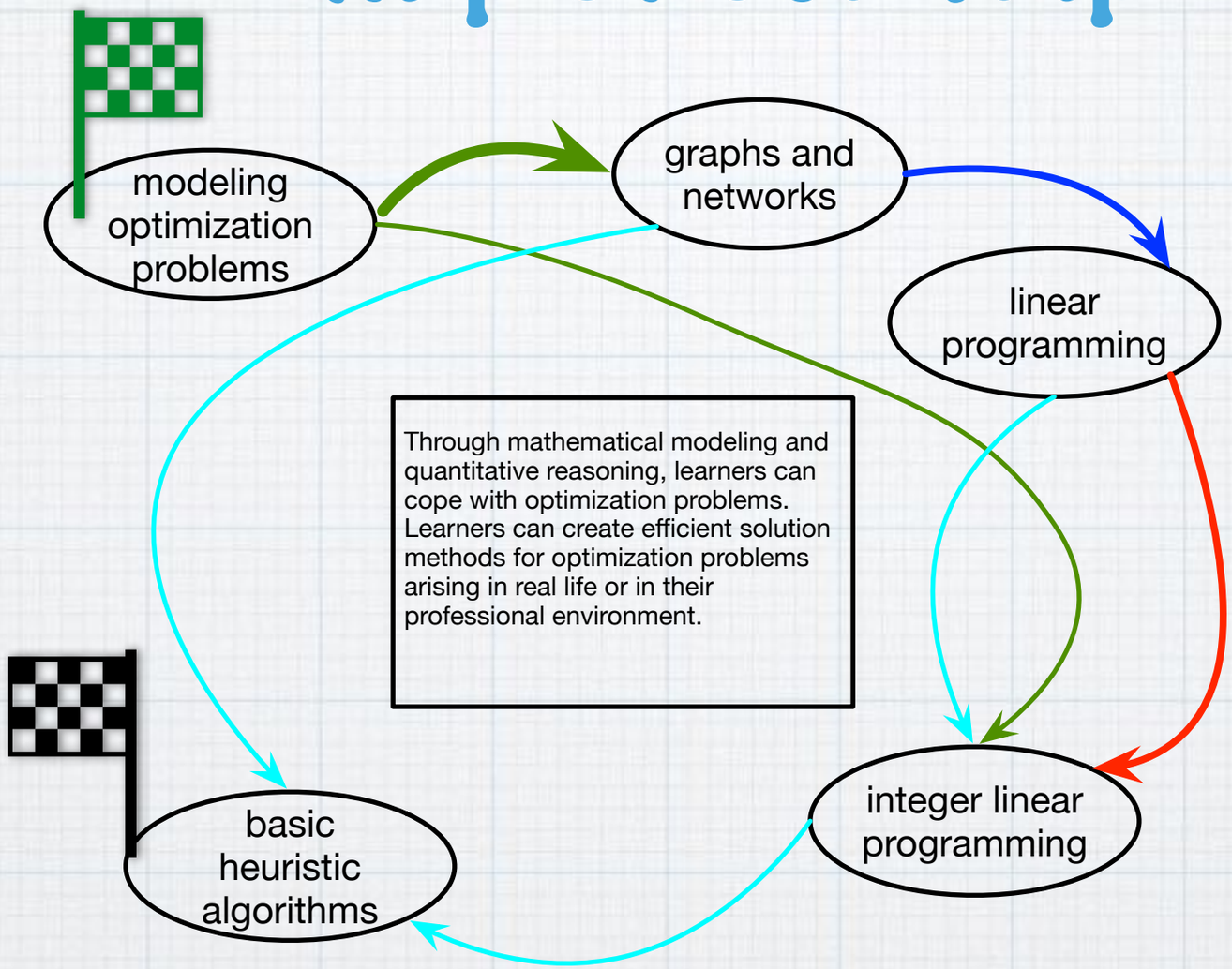
* **Integer Linear Programming and combinatorial optimization**
  Cutting planes methods. Relaxations and branch and bound algorithm.

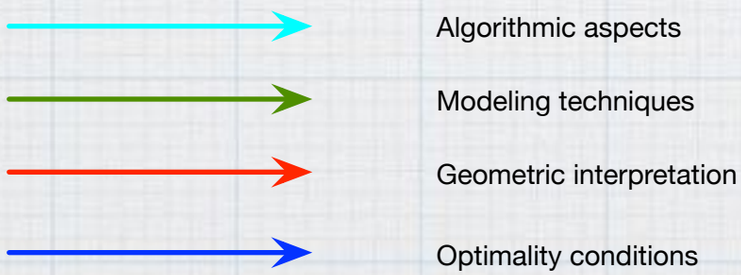
* **Computational complexity and Heuristic algorithms**
  Basics of computational complexity. Greedy and local search.

* **Applications** of OR in computer science and telecommunications: Routing, scheduling, location, transportation.

Metacognitive flows: map of our trip

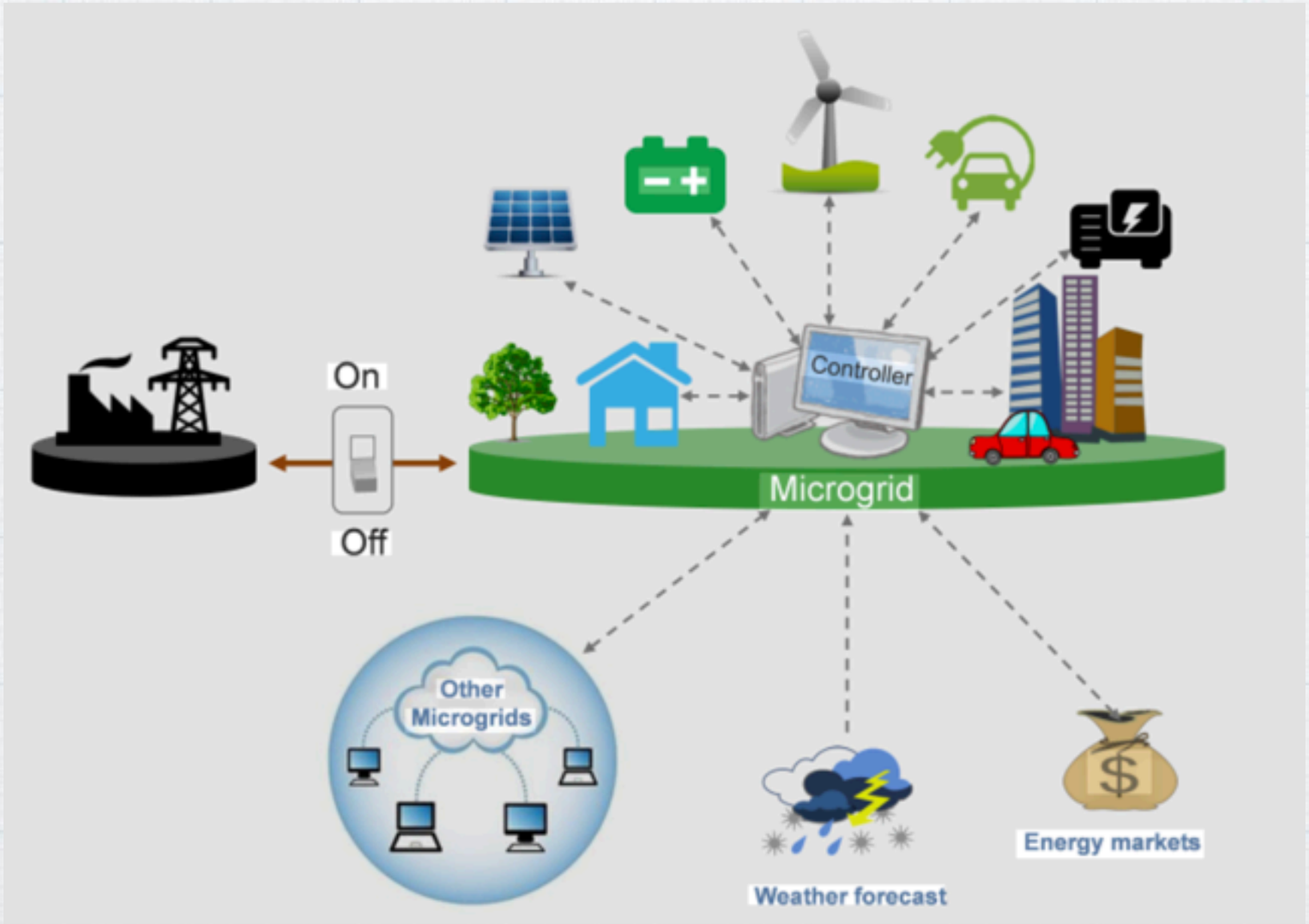






Destination of our trip

Managing a microgrid



Optimizing the use of resources and of the market opportunities

Instructions



Roadmap
Metacognitive instructions



Activity

Resources



In-depth exploration



Your travel journal: self reflections



Call a travel mate



Metacognitive instructions

Motivation of the activity

Scope of the teaching module

It spoils the "surprise" and the pleasure of the discovery

Since you are "adult learners", it is important to let you understand also the metacognitive dimension



Activities

In class pay attention to the logo

mobile app needed Kahoot!

sometimes they will require some preparation at home

1 bonus point (three times) for the best in the ranking

At home

almost weekly I will propose a **puzzle** or a small task to be solved at home

We will discuss the possible solutions together the week after



Activities



Special activity: Peer review

First week of November

Modeling exercise

Individual activity

On average, those participating in the PR score 25% more than the others in the modeling exercise at the exam

Resources



Available on the WeBEEP

- lecture notes in English
- lecture notes in Italian
- slides (**under updating**)
- more than 150 exercises with solutions (**new**)
- exam samples
- lab software
- exercise session texts (in advance)
- solution of the exercises (after)



Some reference books are listed
Some "pills" of OR (under construction)

Travel book and call a travel mate



Learning is also a social experience

Calling a colleague and discussing will be encouraged from time to time



Self reflection helps in constructing your knowledge

From time to time I will ask you to write some reflections on what we have seen

Reading back your reflections can be very instructive

First task: write in your travel journal the name of 2 colleagues to contact

You need help in finding someone, write me an e-mail

Contact me

Your feedback is very important

email: federico.malucelli@polimi.it
begin the subject with "FOR:"

On Teams, under appointment

"OR café"
informal open consultancy

on line Friday at 18:15?
in presence Tuesday at 17:15?
but I'm open to any reasonable timing

The Forum on the WeBEEP



Lab activity

Modeling optimization problems

Solve them with a commercial SW

Python MIP: full version available on the WeBEEP

Advantages: universal, free license even after the course, more flexible

A Lab0 (prelab activity) is available on webeep to get familiar with Python

Exam

- **Written exam : 23 points:**
  - part 1 (13) closed book (!): standard exercises
  - part 2 (10) open book: creative exercises
- **lab exam: 4 points, separate event, one per session**
- **Bonus points** (max 3 + 2) will complete the 4 missing points
  - for those who could not attend the activities there will be an additional question in the first part (4 points)
- **Lab mini projects**: in groups
  - skip the lab exam
- **Sufficiency**: >17.5 in the written exam + lab exam

Active learning or not?

Average grade at the written exam (max 27)

Active students:
22.5



Passive students:
18.4

The active students usually pass the exam in the first two calls

Entry test and survey

The previous knowledge needed in the course is very limited

You are invited to test it

link on webeep

I will give you a general feedback if you need to catch up some concepts

At the beginning of each module I will recall the links with your previous knowledge

I really would like to know something more about yourself

Since the course is mainly on line I've prepared a short anonymous survey

link on webeep

This information will help me in shaping the course on your expectations



https://www.ted.com/talks/marco_lubbecke_anything_you_can_do_i_can_do_better